	of simple cubic functions, the reciprocal function $y = \frac{1}{x}$ with $x \neq 0$, the function $y = k^x$ for integer values of x and simple positive values of k, the trigonometric functions $y = \sin x$ and $y = \cos x$	reciprocal function $y = \frac{1}{x}$ with $x \neq 0$, the exponential function $y = k^x$ for integer values of x and simple positive values of k, the circular functions $y = \sin x$ and $y = \cos x$, within the range -360º to +360º • Recognise the characteristic shapes of all these functions • Draw and plot a range of mathematical functions • Interpret and analyse a range of mathematical functions and be able to draw them, recognising that they were of the correct shape
A q	Construct the graphs of simple loci	• Construct the graphs of simple loci including the circle $x^2 + y^2 = r^2$ for a circle of radius r centred at the origin of the coordinate plane • Find graphically the intersection points of a given straight line with this circle • Select and apply construction techniques and understanding of loci to draw graphs based on circles and perpendiculars of lines

Ref	Content descriptor	Concepts and skills
A r	Construct linear, **quadratic and other** functions from real-life problems and plot their corresponding graphs	• Draw straight line graphs for real-life situations – ready reckoner graphs – conversion graphs – fuel bills – fixed charge (standing charge) and cost per unit – distance time graphs • Draw distance-line graphs • **Generate points and plot graphs of simple quadratic functions, then more general quadratic functions** • **Find approximate solutions of a quadratic equation from the graph of the corresponding quadratic function** • **Find the intersection points of the graphs of a linear and quadratic function, knowing that these are the approximate solutions of the corresponding simultaneous equations representing the linear and quadratic functions**
A s	Discuss, plot and interpret graphs (which may be non-linear) modelling real situations	• Plot a linear graph • Interpret straight line graphs for real-life situations – ready reckoner graphs – conversion graphs – fuel bills – fixed charge (standing charge) and cost per unit • Interpret distance-time graphs • Interpret information presented in a range of linear and non-linear graphs
A t	Generate points and plot graphs of simple quadratic functions, and use these to find approximate solutions	• Generate points and plot graphs of simple quadratic functions, then more general quadratic functions • Find approximate solutions of a quadratic equation from the graph of the corresponding quadratic function • **Select and use the correct mathematical techniques to draw quadratic graphs**

Higher

Ref	Content descriptor	Concepts and skills
A u	**Direct and indirect proportion**	• **Set up and use equations to solve word and other problems involving direct proportion or inverse proportion and relate algebraic solutions to graphical representation of the equations**
A v	**Transformation of functions**	• **Apply to the graph of** $y = \mathrm{f}(x)$ **the transformations** $y = \mathrm{f}(x) + a$**,** $y = \mathrm{f}(ax)$**,** $y = \mathrm{f}(x + a)$**,** $y = a\mathrm{f}(x)$ **for linear, quadratic, sine and cosine functions** $\mathrm{f}(x)$ • **Select and apply the transformations of reflection, rotation, enlargement and translation of functions expressed algebraically** • **Interpret and analyse transformations of functions and write the functions algebraically**

3 Geometry

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM a	Recall and use properties of angles at a point, angles on a straight line (including right angles), perpendicular lines, and opposite angles at a vertex	• Recall and use properties of angles – angles at a point – angles at a point on a straight line, including right angles – perpendicular lines – vertically opposite angles
GM b	Understand and use the angle properties of parallel lines, triangles and quadrilaterals	• Distinguish between scalene, isosceles, equilateral, and right-angled triangles • Understand and use the angle properties of triangles • Use the angle sum of a triangle is 180° • Understand and use the angle properties of intersecting lines • Understand and use the angle properties of parallel lines • Mark parallel lines on a diagram • Use the properties of corresponding and alternate angles • Understand and use the angle properties of quadrilaterals • Give reasons for angle calculations • Explain why the angle sum of a quadrilateral is 360° • Understand the proof that the angle sum of a triangle is 180° • Understand a proof that the exterior angle of a triangle is equal to the sum of the interior angles at the other two vertices • Use the size/angle properties of isosceles and equilateral triangles • **Recall and use these properties of angles in more complex problems**

Ref	Content descriptor	Concepts and skills
GM c	Calculate and use the sums of the interior and exterior angles of polygons	• Calculate and use the sums of the interior angles of polygons • Use geometric language appropriately and recognise and name pentagons, hexagons, heptagons, octagons and decagons • Use the angle sums of irregular polygons • Calculate and use the angles of regular polygons • Use the sum of the interior angles of an n-sided polygon • Use the sum of the exterior angles of any polygon is 360° • Use the sum of the interior angle and the exterior angle is 180° • Find the size of each interior angle or the size of each exterior angle or the number of sides of a regular polygon • Understand tessellations of regular and irregular polygons • Tessellate combinations of polygons • Explain why some shapes tessellate when other shapes do not
GM d	Recall the properties and definitions of special types of quadrilateral, including square, rectangle, parallelogram, trapezium, kite and rhombus	• Recall the properties and definitions of special types of quadrilateral, including symmetry properties • List the properties of each, or identify (name) a given shape • Classify quadrilaterals by their geometric properties
GM e	Recognise reflection and rotation symmetry of 2-D shapes	• Recognise reflection symmetry of 2-D shapes • Identify and draw lines of symmetry on a shape • Recognise rotation symmetry of 2-D shapes • Identify the order of rotational symmetry of a 2-D shape • Draw or complete diagrams with a given number of lines of symmetry • State the line symmetry as a simple algebraic equation • Draw or complete diagrams with a given order of rotational symmetry

Ref	Content descriptor	Concepts and skills
GM f	Understand congruence and similarity	• Recognise that all corresponding angles in similar figures are equal in size when the lengths of sides are not • **Understand and use SSS, SAS, ASA and RHS conditions to prove the congruence of triangles using formal arguments, and to verify standard ruler and a pair of compasses constructions** • **Understand similarity of triangles and of other plane figures, and use this to make geometric inferences** • **Complete a formal geometric proof of similarity of two given triangles**
GM g	Use Pythagoras' theorem in 2-D **and 3-D**	• Understand, recall and use Pythagoras' theorem in 2-D, **then in 3-D problems** • **Understand the language of planes, and recognise the diagonals of a cuboid** • **Calculate the length of a diagonal of a cuboid**
GM h	**Use the trigonometric ratios and the sine and cosine rules to solve 2-D and 3-D problems**	• **Use the trigonometric ratios to solve 2-D and 3-D problems** • **Understand, recall and use trigonometric relationships in right-angled triangles, and use these to solve problems in 2-D and in 3-D configurations** • **Find the angle between a line and a plane (but not the angle between two planes or between two skew lines)** • **Find angles of elevation and angles of depression** • **Use the sine and cosine rules to solve 2-D and 3-D problems**
GM i	Distinguish between centre, radius, chord, diameter, circumference, tangent, arc, sector and segment	• Recall the definition of a circle and identify (name) and draw the parts of a circle • Understand related terms of a circle • Draw a circle given the radius or diameter

Higher

Ref	Content descriptor	Concepts and skills
GM j	**Understand and construct geometrical proofs using circle theorems**	• **Understand and use the fact that the tangent at any point on a circle is perpendicular to the radius at that point** • **Understand and use the fact that tangents from an external point are equal in length** • **Find missing angles on diagrams** • **Give reasons for angle calculations involving the use of tangent theorems** • **Prove and use the facts that:** – **the angle subtended by an arc at the centre of a circle is twice the angle subtended at any point on the circumference** – **the angle in a semicircle is a right angle** – **angles in the same segment are equal** – **opposite angles of a cyclic quadrilateral sum to 180°** – **alternate segment theorem** – **the perpendicular from the centre of a circle to a chord bisect the chord**
GM k	Use 2-D representations of 3-D shapes	• Use 2-D representations of 3-D shapes • Use isometric grids • Draw nets and show how they fold to make a 3-D solid • Understand and draw front and side elevations and plans of shapes made from simple solids • Given the front and side elevations and the plan of a solid, draw a sketch of the 3-D solid

Ref	Content descriptor	Concepts and skills
GM l	Describe and transform 2-D shapes using single or combined rotations, reflections, translations, or enlargements by a positive, **fractional or negative** scale factor and distinguish properties that are preserved under particular transformations	• Describe and transform 2-D shapes using single rotations • Understand that rotations are specified by a centre and an (anticlockwise) angle • Find the centre of rotation • Rotate a shape about the origin, or any other point • Describe and transform 2-D shapes using single reflections • Understand that reflections are specified by a mirror line • Identify the equation of a line of symmetry • Describe and transform 2-D shapes using single translations • Understand that translations are specified by a distance and direction (using a vector) • Translate a given shape by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ • Describe and transform 2-D shapes using enlargements by a positive and **a negative or fractional** scale factor • Understand that an enlargement is specified by a centre and a scale factor • Enlarge shapes using (0, 0) as the centre of enlargement • Enlarge shapes using centre other than (0, 0) • Find the centre of enlargement • Describe and transform 2-D shapes using combined rotations, reflections, translations, or enlargements • Distinguish properties that are preserved under particular transformations • Recognise that enlargements preserve angle but not length

Higher

Ref	Content descriptor	Concepts and skills
GM l	(*Continued*)	• Use congruence to show that translations, rotations and reflections preserve length and angle, so that any figure is congruent to its image under any of these transformations • Understand that distances and angles are preserved under rotations, reflections and translations so that any shape is congruent to its image • Describe a transformation
GM v	Use straight edge and a pair of compasses to carry out constructions	• Use straight edge and a pair of compasses to do standard constructions • Construct a triangle • Construct an equilateral triangle • Understand, from the experience of constructing them, that triangles satisfying SSS, SAS, ASA and RHS are unique, but SSA triangles are not • Construct the perpendicular bisector of a given line • Construct the perpendicular from a point to a line • Construct the perpendicular from a point on a line • Construct the bisector of a given angle • Construct angles of 60°, 90°, 30°, 45° • Draw parallel lines • Draw circles and arcs to a given radius • Construct a regular hexagon inside a circle • Construct diagrams of everyday 2-D situations involving rectangles, triangles, perpendicular and parallel lines • Draw and construct diagrams from given information

Ref	Content descriptor	Concepts and skills
GM w	Construct loci	• Construct: – a region bounded by a circle and an intersecting line – given distance from a point and a given distance from a line – equal distances from two points or two line segments – regions which may be defined by 'nearer to' or 'greater than' • Find and describe regions satisfying a combination of loci (NB: All loci restricted to two dimensions only)
GM x	Calculate perimeters and areas of shapes made from triangles, rectangles **and other shapes**	• Measure shapes to find perimeter or area • Find the perimeter of rectangles and triangles • Calculate perimeter and area of compound shapes made from triangles, rectangles **and other shapes** • Recall and use the formulae for the area of a triangle, rectangle and a parallelogram • Calculate areas of shapes made from triangles and rectangles • Calculate perimeters of compound shapes made from triangles and rectangles • Find the area of a trapezium • Find the area of a parallelogram • Find the surface area of simple shapes (prisms) using the formulae for triangles and rectangles, **and other shapes**
GM y	**Calculate the area of a**	• **Calculate the area of a triangle given the**